

Database Management Systems: Assignment-1

Name: Suraj Shah

Admission Number: I24AI012

Q1. Create a table Student with these fields:

RollNo Number (6), Name Character (25), Semester Number (3), DOB Date,

AdmissionDate Date, HostelRoom Number (5)

Insert 10 Rows in the above created table.

Display the below details:

1. Student table with all columns and rows.

Query:

```
SELECT * FROM student;
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
101	Suraj Shah	4	2001-05-14	2019-07-01	201
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305
103	Ravi Patel	4	2000-11-10	2018-07-01	202
104	Sneha Sharma	2	2003-01-25	2021-07-01	105
105	Ajay Singh	4	2001-08-05	2019-07-01	402
106	Sunil Kumar	1	2004-06-18	2022-07-01	210
107	Anjali Verma	3	2002-12-30	2020-07-01	115
108	Siddharth Jain	4	2001-03-12	2019-07-01	203
109	Rahul Gupta	2	2003-09-05	2021-07-01	312
110	Shivani Desai	4	2000-10-28	2018-07-01	205

2. Student details with columns Roll number and Name only.

Query:

```
SELECT RollNo, Name FROM student;
```

Output:

RollNo	Name
101	Suraj Shah
102	Sakshi Mehta
103	Ravi Patel
104	Sneha Sharma
105	Ajay Singh
106	Sunil Kumar
107	Anjali Verma
108	Siddharth Jain
109	Rahul Gupta
110	Shivani Desai

3. Student details of all the students who are in 4th semester.

Query:

```
SELECT * FROM student WHERE Semester=4;
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
101	Suraj Shah	4	2001-05-14	2019-07-01	201
103	Ravi Patel	4	2000-11-10	2018-07-01	202
105	Ajay Singh	4	2001-08-05	2019-07-01	402
108	Siddharth Jain	4	2001-03-12	2019-07-01	203
110	Shivani Desai	4	2000-10-28	2018-07-01	205

4. Student details of all the students whose roll number is between 100 to 200.

Query:

```
SELECT * FROM student WHERE RollNo BETWEEN 100 AND 200;
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
101	Suraj Shah	4	2001-05-14	2019-07-01	201
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305
103	Ravi Patel	4	2000-11-10	2018-07-01	202
104	Sneha Sharma	2	2003-01-25	2021-07-01	105
105	Ajay Singh	4	2001-08-05	2019-07-01	402
106	Sunil Kumar	1	2004-06-18	2022-07-01	210
107	Anjali Verma	3	2002-12-30	2020-07-01	115
108	Siddharth Jain	4	2001-03-12	2019-07-01	203
109	Rahul Gupta	2	2003-09-05	2021-07-01	312
110	Shivani Desai	4	2000-10-28	2018-07-01	205

5. Student details of all the students whose DOB is greater than 1st jan 2001.

Query:

```
SELECT * FROM student WHERE DOB > '2001-01-01';
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
101	Suraj Shah	4	2001-05-14	2019-07-01	201
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305
104	Sneha Sharma	2	2003-01-25	2021-07-01	105
105	Ajay Singh	4	2001-08-05	2019-07-01	402
106	Sunil Kumar	1	2004-06-18	2022-07-01	210
107	Anjali Verma	3	2002-12-30	2020-07-01	115
108	Siddharth Jain	4	2001-03-12	2019-07-01	203
109	Rahul Gupta	2	2003-09-05	2021-07-01	312

6. Student details of a student with enrollment number '07'

Query:

```
SELECT * FROM student WHERE RollNo=107;
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
107	Anjali Verma	3	2002-12-30	2020-07-01	115

7. Students who reside on the second floor in the hostel.

Query:

```
SELECT * FROM student WHERE HostelRoom LIKE '2%';
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
101	Suraj Shah	4	2001-05-14	2019-07-01	201
103	Ravi Patel	4	2000-11-10	2018-07-01	202
106	Sunil Kumar	1	2004-06-18	2022-07-01	210
108	Siddharth Jain	4	2001-03-12	2019-07-01	203
110	Shivani Desai	4	2000-10-28	2018-07-01	205

8. Students whose name starts with 'S'

Query:

```
SELECT * FROM Student WHERE Name LIKE 'S%';
```

Output:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom
101	Suraj Shah	4	2001-05-14	2019-07-01	201
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305
104	Sneha Sharma	2	2003-01-25	2021-07-01	105
106	Sunil Kumar	1	2004-06-18	2022-07-01	210
108	Siddharth Jain	4	2001-03-12	2019-07-01	203
110	Shivani Desai	4	2000-10-28	2018-07-01	205

Q2. Create a table Employee with these fields:

EmpID Number (6), Name Character (25), Department Character (30), ManagerID

Number (6), JoiningDate Date, Salary Number (8)

Insert 10 Rows in the above created table.

Display the below details:

1. Employee table with all columns and rows.

Query:

```
SELECT * FROM employee;
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary
121	Ravi Kumar	HR	201	2019-03-15	75000
122	Sneha Sharma	Finanace	202	2021-06-10	82000
123	Ajay Singh	IT	203	2018-09-05	95000
124	Anjali Verma	HR	201	2020-02-01	68000
125	Rahul Gupta	Marketing	204	2022-01-20	55000
126	Siddharth Jain	IT	203	2017-11-12	105000
127	Shivani Desai	Finance	202	2020-08-15	72000
128	Sunil Kumar	HR	201	2023-04-12	60000
129	Kavita Mehta	Operations	205	2019-12-18	50000
130	Pooja Patel	IT	203	2021-09-25	88000

2. Employee details with columns Name and Department only.

Query:

```
SELECT Name, Department FROM employee;
```

Output:

Name	Department
Ravi Kumar	HR
Sneha Sharma	Finanace
Ajay Singh	IT
Anjali Verma	HR
Rahul Gupta	Marketing
Siddharth Jain	IT
Shivani Desai	Finance
Sunil Kumar	HR
Kavita Mehta	Operations
Pooja Patel	IT

3. Employee details of all the Employees who are in HR department.

Query:

```
SELECT * FROM employee WHERE Department='HR';
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary
121	Ravi Kumar	HR	201	2019-03-15	75000
124	Anjali Verma	HR	201	2020-02-01	68000
128	Sunil Kumar	HR	201	2023-04-12	60000

4. Employee details of an employee whose EmpID is '124'

Query:

```
SELECT * FROM employee WHERE EmpID=124;
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary
124	Anjali Verma	HR	201	2020-02-01	68000

5. Employee details of all the Employees whose salary is between 50000 to 100000.

Query:

```
SELECT * FROM employee WHERE Salary BETWEEN 50000 AND 100000;
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary
121	Ravi Kumar	HR	201	2019-03-15	75000
122	Sneha Sharma	Finanace	202	2021-06-10	82000
123	Ajay Singh	IT	203	2018-09-05	95000
124	Anjali Verma	HR	201	2020-02-01	68000
125	Rahul Gupta	Marketing	204	2022-01-20	55000
127	Shivani Desai	Finance	202	2020-08-15	72000
128	Sunil Kumar	HR	201	2023-04-12	60000
129	Kavita Mehta	Operations	205	2019-12-18	50000
130	Pooja Patel	IT	203	2021-09-25	88000

6. Employee details of all the Employees whose JoiningDate is greater than 1st jan 2020.

Query:

```
SELECT * FROM employee WHERE JoiningDate > '2020-01-01';
```

Output:

EmpID	Name	Department	ManagerID	JoiningDate	Salary
122	Sneha Sharma	Finanace	202	2021-06-10	82000
124	Anjali Verma	HR	201	2020-02-01	68000
125	Rahul Gupta	Marketing	204	2022-01-20	55000
127	Shivani Desai	Finance	202	2020-08-15	72000
128	Sunil Kumar	HR	201	2023-04-12	60000
130	Pooja Patel	IT	203	2021-09-25	88000